

# SOLIDWORKS API (NET VBA C#) — Reference (260/337): ISimulation- ISimulationDamperFeatureData

SOLIDWORKS API Documentation (converted from CHM)

# ISimulation (Type)

## ISimulation Interface

[See Also](#) [Members](#)



[SolidWorks.Interop.sldworks Namespace](#) : ISimulation Interface

This interface is:

- obsolete and has not been superseded.
- nonfunctional in SOLIDWORKS 2008 and later.

Use the interfaces related to [motion studies](#) introduced in SOLIDWORKS 2008 to access animation and simulation.

**NOTE:** Click the **Members** link, located near the top of the topic, to see this interface's methods and properties.

**.NET Syntax**

Visual Basic (Declaration)

Public Interface ISimulation

Visual Basic (Usage)

Dim instance As [ISimulation](#)

C#

public interface ISimulation

C++/CLI

public interface class ISimulation

**Visual Basic for Applications (VBA) Syntax**

See [Simulation](#).

**Remarks**

Do not confuse Physical Simulation with [animation](#). The SOLIDWORKS Design software computes a Physical Simulation, which generates a number of steps (and transforms) and elapsed time for those steps. The SOLIDWORKS Design software then displays the computed Physical Simulation using animation. To create the display, the animation process takes the Physical Simulator steps and does a linear interpolation of those steps for the elapsed time. The elapsed time and frames of Physical Simulation will most likely be different than the elapsed time and frames of an animation.

**See Also**

[ISimulation Members](#)  
[SolidWorks.Interop.sldworks Namespace](#)

## ISimulation Interface Members

[See Also](#) [Properties](#) [Methods](#)



[SolidWorks.Interop.sldworks Namespace](#) : ISimulation Interface

The following tables list the members exposed by [ISimulation](#).

**Public Properties**

| Name                      | Description               |
|---------------------------|---------------------------|
| <a href="#">Animation</a> | Obsolete. Not superseded. |
| <a href="#">Duration</a>  | Obsolete. Not superseded. |
| <a href="#">Frames</a>    | Obsolete. Not superseded. |
| <a href="#">Status</a>    | Obsolete. Not superseded. |



[TimeStamp](#)

Obsolete. Not superseded.

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## Public Methods

| Name                                                                                                                 | Description               |
|----------------------------------------------------------------------------------------------------------------------|---------------------------|
|  <a href="#">GetFeature</a>         | Obsolete. Not superseded. |
|  <a href="#">IsAnimationPlaying</a> | Obsolete. Not superseded. |
|  <a href="#">PlayAnimation</a>      | Obsolete. Not superseded. |
|  <a href="#">ResetComponents</a>    | Obsolete. Not superseded. |

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## See Also

[ISimulation Interface](#)  
[SolidWorks.Interop.sldworks Namespace](#)

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## ISimulation Interface Methods

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) : ISimulation Interface

For a list of all members of this type, see [ISimulation members](#).

## Public Methods

| Name                                                                                                                   | Description               |
|------------------------------------------------------------------------------------------------------------------------|---------------------------|
|  <a href="#">GetFeature</a>          | Obsolete. Not superseded. |
|  <a href="#">IsAnimationPlaying</a> | Obsolete. Not superseded. |
|  <a href="#">PlayAnimation</a>      | Obsolete. Not superseded. |
|  <a href="#">ResetComponents</a>    | Obsolete. Not superseded. |

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## See Also

[ISimulation Interface](#)  
[SolidWorks.Interop.sldworks Namespace](#)

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## ISimulation Interface Properties

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) : ISimulation Interface

For a list of all members of this type, see [ISimulation members](#).

## Public Properties

| Name                                                                                                          | Description               |
|---------------------------------------------------------------------------------------------------------------|---------------------------|
|  <a href="#">Animation</a> | Obsolete. Not superseded. |
|  <a href="#">Duration</a>  | Obsolete. Not superseded. |
|  <a href="#">Frames</a>    | Obsolete. Not superseded. |
|  <a href="#">Status</a>    | Obsolete. Not superseded. |
|  <a href="#">TimeStamp</a> | Obsolete. Not superseded. |

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## See Also

## Animation Property (ISimulation)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [ISimulation Interface](#) : Animation Property (ISimulation)

Obsolete. Not superseded.

### .NET Syntax

#### Visual Basic (Declaration)

```
ReadOnly Property Animation As Animation
```

#### Visual Basic (Usage)



```
Dim instance As ISimulation  
Dim value As Animation  
  
value = instance.Animation
```

#### C#

```
Animation Animation {get;}
```

#### C++/CLI

```
property Animation^ Animation {  
    Animation^ get();  
}
```

**NOTE:** See [Differences Between Unmanaged C++ and C++/CLI Code](#).

#### Property Value

[Animation](#)

### Visual Basic for Applications (VBA) Syntax

See [Simulation::Animation](#).

### Remarks

| To...                                                  | Then...                                                                       |
|--------------------------------------------------------|-------------------------------------------------------------------------------|
| Get the duration of an animation                       | Use <a href="#">IAnimation::Duration</a> after using this property            |
| Play the animation                                     | Use <a href="#">IAnimation::Play</a> after using this property                |
| Display the <b>Animation Controller</b> pop-up toolbar | Use <a href="#">ISimulation::PlayAnimation</a> instead of using this property |

**NOTE:** Use only the following IAnimation property and method with ISimulation::Animation: IAnimation::Duration and IAnimation::PlayAnimation. The other IAnimation properties and methods do nothing with an IAnimation object returned by ISimulation::Animation because they expect an animation to be playing. Use [ISimulation::IsAnimationPlaying](#) to determine whether an animation is playing.

### See Also

[ISimulation Interface](#)  
[ISimulation Members](#)

### Availability

SOLIDWORKS 2006 FCS, Revision Number 14.0

## Duration Property (ISimulation)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [ISimulation Interface](#) : Duration Property (ISimulation)

Obsolete. Not superseded.

## .NET Syntax

Visual Basic (Declaration)

ReadOnly Property Duration As System.Double

Visual Basic (Usage)



```
Dim instance As ISimulation
Dim value As System.Double

value = instance.Duration
```

C#

```
System.double Duration {get;}
```

C++/CLI

```
property System.double Duration {
    System.double get();
}
```

**NOTE:** See [Differences Between Unmanaged C++ and C++/CLI Code](#).

### Property Value

Elapsed time in seconds

## Visual Basic for Applications (VBA) Syntax

See [Simulation::Duration](#).

## Remarks

If the Physical Simulation has not yet been calculated, then the return value is 0.0.

The return value represents the length of time that the simulation is expected to last. This means that if a bouncing ball takes 15 seconds to come to a stop, then this property returns a value of 15.

## See Also

[ISimulation Interface](#)  
[ISimulation Members](#)

## Availability

SOLIDWORKS 2006 FCS, Revision Number 14.0

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## Frames Property (ISimulation)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [ISimulation Interface](#) : Frames Property (ISimulation)

Obsolete. Not superseded.

## .NET Syntax

Visual Basic (Declaration)

ReadOnly Property Frames As System.Integer

Visual Basic (Usage)



```
Dim instance As ISimulation
Dim value As System.Integer

value = instance.Frames
```

C#

```
System.int Frames {get;}
```

C++/CLI

```
property System.int Frames {
    System.int get();
}
```

**NOTE:** See [Differences Between Unmanaged C++ and C++/CLI Code](#).

#### Property Value

Number of steps

### Visual Basic for Applications (VBA) Syntax

See [Simulation.Frames](#).

### Remarks

If the Physical Simulation is not yet calculated, then the return value is 0.0.

### See Also

[ISimulation Interface](#)  
[ISimulation Members](#)

### Availability

SOLIDWORKS 2006 FCS, Revision Number 14.0

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## GetFeature Method (ISimulation)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [ISimulation Interface](#) : GetFeature Method (ISimulation)

Obsolete. Not superseded.

### .NET Syntax

Visual Basic (Declaration)

```
Function GetFeature() As Feature
```

Visual Basic (Usage)



```
Dim instance As ISimulation
Dim value As Feature

value = instance.GetFeature()
```

C#

```
Feature GetFeature()
```

C++/CLI

```
Feature^ GetFeature();
```

**NOTE:** See [Differences Between Unmanaged C++ and C++/CLI Code](#).

## Return Value

[Feature](#)

## Visual Basic for Applications (VBA) Syntax

See [Simulation::GetFeature](#).

## Remarks

After using this method to get the feature, use either of the following methods to get the [ISimulation](#) interface:

- [IFeature::GetSpecificFeature2](#)
- [IAssemblyDoc::GetSimulation](#)

## See Also

[ISimulation Interface](#)  
[ISimulation Members](#)

## Availability

SOLIDWORKS 2006 FCS, Revision Number 14.0

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## IsAnimationPlaying Method (ISimulation)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [ISimulation Interface](#) : IsAnimationPlaying Method (ISimulation)

Obsolete. Not superseded.

## .NET Syntax

### Visual Basic (Declaration)

```
Function IsAnimationPlaying() As System.Boolean
```

### Visual Basic (Usage)



```
Dim instance As ISimulation  
Dim value As System.Boolean  
  
value = instance.IsAnimationPlaying()
```

### C#

```
System.bool IsAnimationPlaying()
```

### C++/CLI

```
System.bool IsAnimationPlaying();
```

**NOTE:** See [Differences Between Unmanaged C++ and C++/CLI Code](#).

## Return Value

True if an animation is playing, false if not

## Visual Basic for Applications (VBA) Syntax

See [Simulation::IsAnimationPlaying](#).

## See Also

[ISimulation Interface](#)  
[ISimulation Members](#)  
[ISimulation::PlayAnimation Method](#)  
[ISimulation::Animation Property](#)

## Availability

SOLIDWORKS 2006 FCS, Revision Number 14.0

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## PlayAnimation Method (ISimulation)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [ISimulation Interface](#) : PlayAnimation Method (ISimulation)

Obsolete. Not superseded.

## .NET Syntax

Visual Basic (Declaration)

```
Function PlayAnimation() As Animation
```

Visual Basic (Usage)



```
Dim instance As ISimulation  
Dim value As Animation  
  
value = instance.PlayAnimation()
```

C#

```
Animation PlayAnimation()
```

C++/CLI

```
Animation^ PlayAnimation();
```

**NOTE:** See [Differences Between Unmanaged C++ and C++/CLI Code](#).

**Return Value**

[Animation](#)

## Visual Basic for Applications (VBA) Syntax

See [Simulation::PlayAnimation](#).

## Remarks

If an animation is playing when this method is used, then this method returns the [Animation](#) object for that animation. To check to see if an animation is currently playing, use [ISimulation::IsAnimationPlaying](#).

## See Also

[ISimulation Interface](#)  
[ISimulation Members](#)

## Availability

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## ResetComponents Method (ISimulation)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [ISimulation Interface](#) : ResetComponents Method (ISimulation)

Obsolete. Not superseded.

## .NET Syntax

Visual Basic (Declaration)



## Function ResetComponents() As System.Boolean

Visual Basic (Usage)



```
Dim instance As ISimulation  
Dim value As System.Boolean
```

```
value = instance.ResetComponents()
```

C#

```
System.bool ResetComponents()
```

C++/CLI

```
System.bool ResetComponents();
```

**NOTE:** See [Differences Between Unmanaged C++ and C++/CLI Code](#).

### Return Value

True if the components' positions and orientations are reset, false if not

## Visual Basic for Applications (VBA) Syntax

See [Simulation::ResetComponents](#).

## See Also

[ISimulation Interface](#)  
[ISimulation Members](#)

## Availability

SOLIDWORKS 2006 FCS, Revision Number 14.0

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## Status Property (ISimulation)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [ISimulation Interface](#) : Status Property (ISimulation)

Obsolete. Not superseded.

## .NET Syntax

Visual Basic (Declaration)

```
ReadOnly Property Status As System.Boolean
```

Visual Basic (Usage)



```
Dim instance As ISimulation  
Dim value As System.Boolean
```

```
value = instance.Status
```

C#

```
System.bool Status {get;}
```

C++/CLI

```
property System.bool Status {  
    System.bool get();  
}
```

**NOTE:** See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Property Value

True if the Physical Simulation has been calculated, false if it has not or if the calculation was unsuccessful

Visual Basic for Applications (VBA) Syntax

See [Simulation::Status](#).

See Also

[ISimulation Interface](#)  
[ISimulation Members](#)

Availability

SOLIDWORKS 2006 FCS, Revision Number 14.0

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TimeStamp Property (ISimulation)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [ISimulation Interface](#) : TimeStamp Property (ISimulation)

Obsolete. Not superseded.

.NET Syntax

Visual Basic (Declaration)

ReadOnly Property TimeStamp As System.Integer

Visual Basic (Usage)



```
Dim instance As ISimulation  
Dim value As System.Integer  
  
value = instance.TimeStamp
```

C#

```
System.int TimeStamp {get;}
```

C++/CLI

```
property System.int TimeStamp {  
    System.int get();  
}
```

**NOTE:** See [Differences Between Unmanaged C++ and C++/CLI Code](#).

Property Value

Time stamp for this Physical Simulation (see **Remarks**)

Visual Basic for Applications (VBA) Syntax

See [Simulation::Timestamp](#).

Remarks

| If a...                                                          | Then...                               |
|------------------------------------------------------------------|---------------------------------------|
| New Physical Simulation is calculated                            | The time stamp (return value) changes |
| Physical Simulation does not exist or it was deleted             | 0 is returned                         |
| Physical Simulation was created using SOLIDWORKS 2005 or earlier | -1 is returned                        |

## See Also

[ISimulation Interface](#)  
[ISimulation Members](#)

## Availability

SOLIDWORKS 2006 FCS, Revision Number 14.0

# ISimulation3DContactFeatureData (Type)

## ISimulation3DContactFeatureData Interface

[See Also](#) [Members](#) [Example](#)



[SolidWorks.Interop.sldworks Namespace](#) : ISimulation3DContactFeatureData Interface

Allows access to a 3D Contact feature in SOLIDWORKS Motion studies.

**NOTE:** Click the **Members** link, located near the top of the topic, to see this interface's methods and properties.

### .NET Syntax

Visual Basic (Declaration)

```
Public Interface ISimulation3DContactFeatureData
```

Visual Basic (Usage)



```
Dim instance As ISimulation3DContactFeatureData
```

C#

```
public interface ISimulation3DContactFeatureData
```

C++/CLI

```
public interface class ISimulation3DContactFeatureData
```

### Visual Basic for Applications (VBA) Syntax

See [Simulation3DContactFeatureData](#).

### Example

[Create 3D Contact Feature \(VBA\)](#)  
[Create 3D Contact Feature \(VB.NET\)](#)  
[Create 3D Contact Feature \(C#\)](#)

### Accessors

[IFeature::GetDefinition](#) and [IFeature::IGetDefintion](#)  
[IMotionStudy::CreateDefinition](#)

### Access Diagram

[Simulation3DContactFeatureData](#)

### See Also

[ISimulation3DContactFeatureData Members](#)  
[SolidWorks.Interop.sldworks Namespace](#)  
[ISimulationDamperFeatureData Interface](#)  
[ISimulationForceFeatureData Interface](#)  
[ISimulationGravityFeatureData Interface](#)  
[ISimulationMotorFeatureData Interface](#)  
[ISimulationSpringFeatureData Interface](#)

## ISimulation3DContactFeatureData Interface Members

[See Also](#) [Properties](#)



[SolidWorks.Interop.sldworks Namespace](#) : ISimulation3DContactFeatureData Interface

The following tables list the members exposed by [ISimulation3DContactFeatureData](#).

### Public Properties

| Name                                                                                                                         | Description                                                                                                                                                |
|------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------|
|  <a href="#">ContactComponents</a>          | Gets or sets the components to check for contact in a 3D Contact feature.                                                                                  |
|  <a href="#">DynamicFrictionCoefficient</a> | Gets or sets the dynamic friction coefficient in a 3D Contact feature.                                                                                     |
|  <a href="#">DynamicFrictionVelocity</a>    | Gets or sets the dynamic friction velocity in a 3D Contact feature.                                                                                        |
|  <a href="#">Exponent</a>                   | Gets or sets the exponent used to calculate the exponential force in a 3D Contact feature.                                                                 |
|  <a href="#">FrictionOption</a>             | Gets or sets whether contact friction is off, full (static), or dynamic in a 3D Contact feature.                                                           |
|  <a href="#">MaterialID</a>                 | Gets or sets the type of material the specified component in this 3D Contact feature.                                                                      |
|  <a href="#">MaxDamping</a>                 | Gets or sets the maximum damping coefficient of the boundary of interaction in a 3D Contact feature.                                                       |
|  <a href="#">Penetration</a>                | Gets or sets the interaction boundary penetration at which the maximum damping value is applied when calculating the impact force in a 3D Contact feature. |
|  <a href="#">RestitutionCoefficient</a>     | Gets or sets the coefficient of restitution in a 3D Contact feature.                                                                                       |
|  <a href="#">SpecifyMaterial</a>            | Gets or sets whether to use materials in a 3D Contact feature.                                                                                             |
|  <a href="#">StaticFrictionCoefficient</a>  | Gets or sets the static friction coefficient in a 3D Contact feature.                                                                                      |
|  <a href="#">StaticFrictionVelocity</a>     | Gets or sets the static friction velocity in a 3D Contact feature.                                                                                         |
|  <a href="#">Stiffness</a>                  | Gets or sets the stiffness at the boundary of interaction between the two parts in a 3D Contact feature.                                                   |
|  <a href="#">UseRestitutionCoefficient</a>  | Gets or sets whether to use the restitution coefficient instead of impact force in a 3D Contact feature.                                                   |

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## See Also

[ISimulation3DContactFeatureData Interface](#)  
[SolidWorks.Interop.sldworks Namespace](#)  
[ISimulationDamperFeatureData Interface](#)  
[ISimulationForceFeatureData Interface](#)  
[ISimulationGravityFeatureData Interface](#)  
[ISimulationMotorFeatureData Interface](#)  
[ISimulationSpringFeatureData Interface](#)

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## ISimulation3DContactFeatureData Interface Properties

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) : ISimulation3DContactFeatureData Interface

For a list of all members of this type, see [ISimulation3DContactFeatureData members](#).

## Public Properties

| Name                                                                                                                           | Description                                                                                                                                                |
|--------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------|
|  <a href="#">ContactComponents</a>          | Gets or sets the components to check for contact in a 3D Contact feature.                                                                                  |
|  <a href="#">DynamicFrictionCoefficient</a> | Gets or sets the dynamic friction coefficient in a 3D Contact feature.                                                                                     |
|  <a href="#">DynamicFrictionVelocity</a>    | Gets or sets the dynamic friction velocity in a 3D Contact feature.                                                                                        |
|  <a href="#">Exponent</a>                   | Gets or sets the exponent used to calculate the exponential force in a 3D Contact feature.                                                                 |
|  <a href="#">FrictionOption</a>             | Gets or sets whether contact friction is off, full (static), or dynamic in a 3D Contact feature.                                                           |
|  <a href="#">MaterialID</a>                 | Gets or sets the type of material the specified component in this 3D Contact feature.                                                                      |
|  <a href="#">MaxDamping</a>                 | Gets or sets the maximum damping coefficient of the boundary of interaction in a 3D Contact feature.                                                       |
|  <a href="#">Penetration</a>                | Gets or sets the interaction boundary penetration at which the maximum damping value is applied when calculating the impact force in a 3D Contact feature. |
|  <a href="#">RestitutionCoefficient</a>     | Gets or sets the coefficient of restitution in a 3D Contact feature.                                                                                       |
|  <a href="#">SpecifyMaterial</a>            | Gets or sets whether to use materials in a 3D Contact feature.                                                                                             |
|  <a href="#">StaticFrictionCoefficient</a>  | Gets or sets the static friction coefficient in a 3D Contact feature.                                                                                      |
|  <a href="#">StaticFrictionVelocity</a>     | Gets or sets the static friction velocity in a 3D Contact feature.                                                                                         |
|  <a href="#">Stiffness</a>                  | Gets or sets the stiffness at the boundary of interaction between the two parts in a 3D Contact feature.                                                   |
|  <a href="#">UseRestitutionCoefficient</a>  | Gets or sets whether to use the restitution coefficient instead of impact force in a 3D Contact feature.                                                   |

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## See Also

[ISimulation3DContactFeatureData Interface](#)  
[SolidWorks.Interop.sldworks Namespace](#)  
[ISimulationDamperFeatureData Interface](#)  
[ISimulationForceFeatureData Interface](#)  
[ISimulationGravityFeatureData Interface](#)  
[ISimulationMotorFeatureData Interface](#)  
[ISimulationSpringFeatureData Interface](#)

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## ContactComponents Property (ISimulation3DContactFeatureData)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [ISimulation3DContactFeatureData Interface](#) : ContactComponents Property (ISimulation3DContactFeatureData)

Gets or sets the components to check for contact in a 3D Contact feature.

## .NET Syntax

### Visual Basic (Declaration)

Property ContactComponents As System.Object

### Visual Basic (Usage)



```
Dim instance As ISimulation3DContactFeatureData  
Dim value As System.Object
```

```
instance.ContactComponents = value
```

```
value = instance.ContactComponents
```

### C#

```
System.object ContactComponents {get; set;}
```

### C++/CLI

```
property System.Object^ ContactComponents {  
    System.Object^ get();  
    void set ( & System.Object^ value);  
}
```

**NOTE:** See [Differences Between Unmanaged C++ and C++/CLI Code](#).

### Property Value

Array of [components](#)

## Visual Basic for Applications (VBA) Syntax

See [Simulation3DContactFeatureData::ContactComponents](#).

## Example

See the [ISimulation3DContactFeatureData](#) examples.

## See Also

[ISimulation3DContactFeatureData Interface](#)  
[ISimulation3DContactFeatureData Members](#)

## Availability

SOLIDWORKS 2009 FCS, Revision Number 17.0

SOLIDWORKS API Help

[Send comments on this topic.](#)

## DynamicFrictionCoefficient Property (ISimulation3DContactFeatureData)

[See Also](#)



Gets or sets the dynamic friction coefficient in a 3D Contact feature.

## .NET Syntax

### Visual Basic (Declaration)

Property DynamicFrictionCoefficient As System.Double

### Visual Basic (Usage)



```
Dim instance As ISimulation3DContactFeatureData  
Dim value As System.Double
```

```
instance.DynamicFrictionCoefficient = value
```

```
value = instance.DynamicFrictionCoefficient
```

### C#

```
System.double DynamicFrictionCoefficient {get; set;}
```

### C++/CLI

```
property System.double DynamicFrictionCoefficient {  
    System.double get();  
    void set ( & System.double value);  
}
```

**NOTE:** See [Differences Between Unmanaged C++ and C++/CLI Code](#).

### Property Value

Dynamic friction coefficient

## Visual Basic for Applications (VBA) Syntax

See [Simulation3DContactFeatureData::DynamicFrictionCoefficient](#).

## Example

See the [ISimulation3DContactFeatureData](#) examples.

## Remarks

Available only when [ISimulation3DContactFeatureData::FrictionOption](#) is [swMotionContactFrictionType\\_e.swMotionContactFrictionDynamic](#).

## See Also

[ISimulation3DContactFeatureData Interface](#)  
[ISimulation3DContactFeatureData Members](#)  
[ISimulation3DContactFeatureData::DynamicFrictionVelocity Property](#)

## Availability

SOLIDWORKS 2009 FCS, Revision Number 17.0

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## DynamicFrictionVelocity Property (ISimulation3DContactFeatureData)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [ISimulation3DContactFeatureData Interface](#) : DynamicFrictionVelocity Property (ISimulation3DContactFeatureData)

Gets or sets the dynamic friction velocity in a 3D Contact feature.

## .NET Syntax

### Visual Basic (Declaration)

## Property DynamicFrictionVelocity As System.Double

Visual Basic (Usage)



```
Dim instance As ISimulation3DContactFeatureData  
Dim value As System.Double  
  
instance.DynamicFrictionVelocity = value  
value = instance.DynamicFrictionVelocity
```

C#

```
System.double DynamicFrictionVelocity {get; set;}
```

C++/CLI

```
property System.double DynamicFrictionVelocity {  
    System.double get();  
    void set ( &    System.double value);  
}
```

**NOTE:** See [Differences Between Unmanaged C++ and C++/CLI Code](#).

### Property Value

Dynamic friction velocity

### Visual Basic for Applications (VBA) Syntax

See [Simulation3DContactFeatureData::DynamicFrictionVelocity](#).

### Example

See the [ISimulation3DContactFeatureData](#) examples.

### Remarks

Available only when [ISimulation3DContactFeatureData::FrictionOption](#) is [swMotionContactFrictionType\\_e.swMotionContactFrictionDynamic](#).

### See Also

[ISimulation3DContactFeatureData Interface](#)  
[ISimulation3DContactFeatureData Members](#)  
[ISimulation3DContactFeatureData::DynamicFrictionCoefficient Property](#)

### Availability

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## Exponent Property (ISimulation3DContactFeatureData)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [ISimulation3DContactFeatureData Interface](#) : Exponent Property (ISimulation3DContactFeatureData)

Gets or sets the exponent used to calculate the exponential force in a 3D Contact feature.

### .NET Syntax

Visual Basic (Declaration)

Property Exponent As System.Double

Visual Basic (Usage)



```
Dim instance As ISimulation3DContactFeatureData  
Dim value As System.Double
```



```
instance.Exponent = value  
value = instance.Exponent
```

C#

```
System.double Exponent {get; set;}
```

C++/CLI

```
property System.double Exponent {  
    System.double get();  
    void set ( & System.double value);  
}
```

**NOTE:** See [Differences Between Unmanaged C++ and C++/CLI Code](#).

#### Property Value

Exponent

### Visual Basic for Applications (VBA) Syntax

See [Simulation3DContactFeatureData::Exponent](#).

### Example

See the [ISimulation3DContactFeatureData](#) examples.

### Remarks

Available only when [ISimulation3DContactFeatureData::UseRestitutionCoefficient](#) and [ISimulation3DContactFeatureData::SpecifyMaterial](#) are both false.

### See Also

[ISimulation3DContactFeatureData Interface](#)  
[ISimulation3DContactFeatureData Members](#)

### Availability

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## FrictionOption Property (ISimulation3DContactFeatureData)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [ISimulation3DContactFeatureData Interface](#) : FrictionOption Property (ISimulation3DContactFeatureData)

Gets or sets whether contact friction is off, full (static), or dynamic in a 3D Contact feature.

### .NET Syntax

Visual Basic (Declaration)

```
Property FrictionOption As System.Integer
```

Visual Basic (Usage)



```
Dim instance As ISimulation3DContactFeatureData  
Dim value As System.Integer
```

```
instance.FrictionOption = value  
value = instance.FrictionOption
```

C#

```
System.int FrictionOption {get; set;}
```

C++/CLI

```
property System.int FrictionOption {
    System.int get();
    void set ( & System.int value);
}
```

**NOTE:** See [Differences Between Unmanaged C++ and C++/CLI Code](#).

#### Property Value

Friction as defined in [swMotionContactFrictionType\\_e](#)

### Visual Basic for Applications (VBA) Syntax

See [Simulation3DContactFeatureData::FrictionOption](#).

### Example

See the [ISimulation3DContactFeatureData](#) examples.

### See Also

[ISimulation3DContactFeatureData Interface](#)  
[ISimulation3DContactFeatureData Members](#)  
[ISimulation3DContactFeatureData::DynamicFrictionCoefficient Property](#)  
[ISimulation3DContactFeatureData::DynamicFrictionVelocity Property](#)  
[ISimulation3DContactFeatureData::StaticFrictionCoefficient Property](#)  
[ISimulation3DContactFeatureData::StaticFrictionVelocity Property](#)

### Availability

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## MaterialID Property (ISimulation3DContactFeatureData)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [ISimulation3DContactFeatureData Interface](#) : MaterialID Property (ISimulation3DContactFeatureData)

Gets or sets the type of material the specified component in this 3D Contact feature.

### .NET Syntax

Visual Basic (Declaration)

```
Property MaterialID(
    ByVal WhichOne As System.Integer _
) As System.Integer
```

Visual Basic (Usage)



```
Dim instance As ISimulation3DContactFeatureData
Dim WhichOne As System.Integer
Dim value As System.Integer
```

```
instance.MaterialID(WhichOne) = value
```

```
value = instance.MaterialID(WhichOne)
```

C#

```
System.int MaterialID(
    System.int WhichOne
) {get; set;}
```

C++/CLI

```
property System.int MaterialID {
    System.int get(System.int WhichOne);
```

```
void set (System.int WhichOne, System.int value);  
}
```

**NOTE:** See [Differences Between Unmanaged C++ and C++/CLI Code](#).

#### Parameters

*WhichOne*

0-based index of the contact component

#### Property Value

Material as defined by [swCosmosWorksMat](#)

### Visual Basic for Applications (VBA) Syntax

See [Simulation3DContactFeatureData::MaterialID](#).

### Example

See the [ISimulation3DContactFeatureData](#) examples.

### Remarks

Available only when [ISimulation3DContactFeatureData::SpecifyMaterial](#) is true.

### See Also

[ISimulation3DContactFeatureData Interface](#)  
[ISimulation3DContactFeatureData Members](#)

### Availability

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## MaxDamping Property (ISimulation3DContactFeatureData)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [ISimulation3DContactFeatureData Interface](#) : MaxDamping Property (ISimulation3DContactFeatureData)

Gets or sets the maximum damping coefficient of the boundary of interaction in a 3D Contact feature.

### .NET Syntax

Visual Basic (Declaration)

Property MaxDamping As System.Double

Visual Basic (Usage)



```
Dim instance As ISimulation3DContactFeatureData  
Dim value As System.Double
```

```
instance.MaxDamping = value
```

```
value = instance.MaxDamping
```

C#

```
System.double MaxDamping {get; set;}
```

C++/CLI

```
property System.double MaxDamping {  
    System.double get();  
    void set ( & System.double value);  
}
```

**NOTE:** See [Differences Between Unmanaged C++ and C++/CLI Code](#).

### Property Value

Maximum damping coefficient of the boundary of interaction

### Visual Basic for Applications (VBA) Syntax

See [Simulation3DContactFeatureData::MaxDamping](#).

### Example

See the [ISimulation3DContactFeatureData](#) examples.

### Remarks

Available only when [ISimulation3DContactFeatureData::UseRestitutionCoefficient](#) and [ISimulation3DContactFeatureData::SpecifyMaterial](#) are both false.

### See Also

[ISimulation3DContactFeatureData Interface](#)  
[ISimulation3DContactFeatureData Members](#)

### Availability

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## Penetration Property (ISimulation3DContactFeatureData)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [ISimulation3DContactFeatureData Interface](#) : Penetration Property (ISimulation3DContactFeatureData)

Gets or sets the interaction boundary penetration at which the maximum damping value is applied when calculating the impact force in a 3D Contact feature.

### .NET Syntax

#### Visual Basic (Declaration)

Property Penetration As System.Double

#### Visual Basic (Usage)



```
Dim instance As ISimulation3DContactFeatureData  
Dim value As System.Double
```

```
instance.Penetration = value
```

```
value = instance.Penetration
```

#### C#

```
System.double Penetration {get; set;}
```

#### C++/CLI

```
property System.double Penetration {  
    System.double get();  
    void set ( & System.double value);  
}
```

**NOTE:** See [Differences Between Unmanaged C++ and C++/CLI Code](#).

### Property Value

Penetration in mechanism's length unit

### Visual Basic for Applications (VBA) Syntax

See [Simulation3DContactFeatureData::Penetration](#).

## Example

See the [ISimulation3DContactFeatureData](#) examples.

## Remarks

Available only when [ISimulation3DContactFeatureData::UseRestitutionCoefficient](#) and [ISimulation3DContactFeatureData::SpecifyMaterial](#) are both false.

## See Also

[ISimulation3DContactFeatureData Interface](#)  
[ISimulation3DContactFeatureData Members](#)

## Availability

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## RestitutionCoefficient Property (ISimulation3DContactFeatureData)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [ISimulation3DContactFeatureData Interface](#) : RestitutionCoefficient Property (ISimulation3DContactFeatureData)

Gets or sets the coefficient of restitution in a 3D Contact feature.

## .NET Syntax

Visual Basic (Declaration)

Property RestitutionCoefficient As System.Double

Visual Basic (Usage)



```
Dim instance As ISimulation3DContactFeatureData
Dim value As System.Double

instance.RestitutionCoefficient = value
value = instance.RestitutionCoefficient
```

C#

```
System.double RestitutionCoefficient {get; set;}
```

C++/CLI

```
property System.double RestitutionCoefficient {
    System.double get();
    void set ( & System.double value);
}
```

**NOTE:** See [Differences Between Unmanaged C++ and C++/CLI Code](#).

### Property Value

Coefficient of restitution

## Visual Basic for Applications (VBA) Syntax

See [Simulation3DContactFeatureData::RestitutionCoefficient](#).

## Example

See the [ISimulation3DContactFeatureData](#) examples.

## Remarks

Available only when [ISimulation3DContactFeatureData::UseRestitutionCoefficient](#) is true.

The restitution coefficient is the ratio of relative velocities of two elastic spheres before and after impact.

## See Also

[ISimulation3DContactFeatureData Interface](#)  
[ISimulation3DContactFeatureData Members](#)

## Availability

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## SpecifyMaterial Property (ISimulation3DContactFeatureData)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [ISimulation3DContactFeatureData Interface](#) : SpecifyMaterial Property (ISimulation3DContactFeatureData)

Gets or sets whether to use materials in a 3D Contact feature.

## .NET Syntax

Visual Basic (Declaration)

Property SpecifyMaterial As System.Boolean

Visual Basic (Usage)



```
Dim instance As ISimulation3DContactFeatureData  
Dim value As System.Boolean
```

```
instance.SpecifyMaterial = value
```

```
value = instance.SpecifyMaterial
```

C#

```
System.bool SpecifyMaterial {get; set;}
```

C++/CLI

```
property System.bool SpecifyMaterial {  
    System.bool get();  
    void set ( & System.bool value);  
}
```

**NOTE:** See [Differences Between Unmanaged C++ and C++/CLI Code](#).

### Property Value

True to use materials, false to not

## Visual Basic for Applications (VBA) Syntax

See [Simulation3DContactFeatureData::SpecifyMaterial](#).

## Example

See the [ISimulation3DContactFeatureData](#) examples.

## Remarks

Set to false to modify elastic properties:

- [ISimulation3DContactFeatureData::Stiffness](#)
- [ISimulation3DContactFeatureData::Exponent](#)
- [ISimulation3DContactFeatureData::MaxDamping](#)
- [ISimulation3DContactFeatureData::Penetration](#)

## See Also

[ISimulation3DContactFeatureData Interface](#)  
[ISimulation3DContactFeatureData Members](#)  
[ISimulation3DContactFeatureData::MaterialID Property](#)

## Availability

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## StaticFrictionCoefficient Property (ISimulation3DContactFeatureData)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [ISimulation3DContactFeatureData Interface](#) : StaticFrictionCoefficient Property (ISimulation3DContactFeatureData)

Gets or sets the static friction coefficient in a 3D Contact feature.

## .NET Syntax

### Visual Basic (Declaration)

Property StaticFrictionCoefficient As System.Double

### Visual Basic (Usage)



```
Dim instance As ISimulation3DContactFeatureData  
Dim value As System.Double
```

```
instance.StaticFrictionCoefficient = value
```

```
value = instance.StaticFrictionCoefficient
```

### C#

```
System.double StaticFrictionCoefficient {get; set;}
```

### C++/CLI

```
property System.double StaticFrictionCoefficient {  
    System.double get();  
    void set ( & System.double value);  
}
```

**NOTE:** See [Differences Between Unmanaged C++ and C++/CLI Code](#).

### Property Value

Static friction coefficient

## Visual Basic for Applications (VBA) Syntax

See [Simulation3DContactFeatureData::StaticFrictionCoefficient](#).

## Example

See the [ISimulation3DContactFeatureData](#) examples.

## Remarks

Available only when [ISimulation3DContactFeatureData::FrictionOption](#) is [swMotionContactFrictionType\\_e.swMotionContactFrictionFull](#).

## See Also

[ISimulation3DContactFeatureData Interface](#)  
[ISimulation3DContactFeatureData Members](#)  
[ISimulation3DContactFeatureData::StaticFrictionVelocity Property](#)

## Availability

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## StaticFrictionVelocity Property (ISimulation3DContactFeatureData)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [ISimulation3DContactFeatureData Interface](#) : StaticFrictionVelocity Property (ISimulation3DContactFeatureData)

Gets or sets the static friction velocity in a 3D Contact feature.

## Visual Basic Syntax

Visual Basic (Declaration)

Property StaticFrictionVelocity As System.Double

Visual Basic (Usage)



```
Dim instance As ISimulation3DContactFeatureData  
Dim value As System.Double
```

```
instance.StaticFrictionVelocity = value
```

```
value = instance.StaticFrictionVelocity
```

C#

```
System.double StaticFrictionVelocity {get; set;}
```

C++/CLI

```
property System.double StaticFrictionVelocity {  
    System.double get();  
    void set ( & System.double value);  
}
```

**NOTE:** See [Differences Between Unmanaged C++ and C++/CLI Code](#).

### Property Value

Static friction velocity

## Visual Basic for Applications (VBA) Syntax

See [Simulation3DContactFeatureData::StaticFrictionVelocity](#).

## Example

See the [ISimulation3DContactFeatureData](#) examples.

## Remarks

Available only when [ISimulation3DContactFeatureData::FrictionOption](#) is [swMotionContactFrictionType\\_e.swMotionContactFrictionFull](#).

## See Also

[ISimulation3DContactFeatureData Interface](#)

[ISimulation3DContactFeatureData Members](#)

[ISimulation3DContactFeatureData::StaticFrictionCoefficient Property](#)

## Availability

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## Stiffness Property (ISimulation3DContactFeatureData)

[See Also](#)





[SolidWorks.Interop.sldworks Namespace](#) > [ISimulation3DContactFeatureData Interface](#) : Stiffness Property (ISimulation3DContactFeatureData)

Gets or sets the stiffness at the boundary of interaction between the two parts in a 3D Contact feature.

## .NET Syntax

### Visual Basic (Declaration)

Property Stiffness As System.Double

### Visual Basic (Usage)



```
Dim instance As ISimulation3DContactFeatureData  
Dim value As System.Double
```

```
instance.Stiffness = value
```

```
value = instance.Stiffness
```

### C#

```
System.double Stiffness {get; set;}
```

### C++/CLI

```
property System.double Stiffness {  
    System.double get();  
    void set ( & System.double value);  
}
```

**NOTE:** See [Differences Between Unmanaged C++ and C++/CLI Code](#).

### Property Value

Stiffness at the boundary of interaction

## Visual Basic for Applications (VBA) Syntax

See [Simulation3DContactFeatureData::Stiffness](#).

## Example

See the [ISimulation3DContactFeatureData](#) examples.

## Remarks

Available only when [ISimulation3DContactFeatureData::UseRestitutionCoefficient](#) and [ISimulation3DContactFeatureData::SpecifyMaterial](#) are both false.

## See Also

[ISimulation3DContactFeatureData Interface](#)  
[ISimulation3DContactFeatureData Members](#)

## Availability

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## UseRestitutionCoefficient Property (ISimulation3DContactFeatureData)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [ISimulation3DContactFeatureData Interface](#) : UseRestitutionCoefficient Property (ISimulation3DContactFeatureData)

Gets or sets whether to use the restitution coefficient instead of impact force in a 3D Contact feature.

## .NET Syntax

### Visual Basic (Declaration)

## Property UseRestitutionCoefficient As System.Boolean

Visual Basic (Usage)



```
Dim instance As ISimulation3DContactFeatureData  
Dim value As System.Boolean
```

```
instance.UseRestitutionCoefficient = value
```

```
value = instance.UseRestitutionCoefficient
```

C#

```
System.bool UseRestitutionCoefficient {get; set;}
```

C++/CLI

```
property System.bool UseRestitutionCoefficient {  
    System.bool get();  
    void set ( & System.bool value);  
}
```

**NOTE:** See [Differences Between Unmanaged C++ and C++/CLI Code](#).

### Property Value

True to use the restitution coefficient, false to use impact force

## Visual Basic for Applications (VBA) Syntax

See [Simulation3DContactFeatureData::UseRestitutionCoefficient](#).

## Example

See the [ISimulation3DContactFeatureData](#) examples.

## See Also

[ISimulation3DContactFeatureData Interface](#)  
[ISimulation3DContactFeatureData Members](#)  
[ISimulation3DContactFeatureData::MaxDamping Property](#)  
[ISimulation3DContactFeatureData::Exponent Property](#)  
[ISimulation3DContactFeatureData::Penetration Property](#)  
[ISimulation3DContactFeatureData::RestitutionCoefficient Property](#)  
[ISimulation3DContactFeatureData::Stiffness Property](#)

## Availability

SOLIDWORKS 2009 FCS, Revision Number 17.0

# ISimulationDamperFeatureData (Type)

## ISimulationDamperFeatureData Interface

[See Also](#) [Members](#) [Example](#)



[SolidWorks.Interop.sldworks Namespace](#) : ISimulationDamperFeatureData Interface

Allows access to a damper feature in SOLIDWORKS Motion studies.  
**NOTE:** Click the **Members** link, located near the top of the topic, to see this interface's methods and properties.

### .NET Syntax

Visual Basic (Declaration)

```
Public Interface ISimulationDamperFeatureData
```

Visual Basic (Usage)



```
Dim instance As ISimulationDamperFeatureData
```

C#

```
public interface ISimulationDamperFeatureData
```

C++/CLI

```
public interface class ISimulationDamperFeatureData
```

### Visual Basic for Applications (VBA) Syntax

See [SimulationDamperFeatureData](#).

### Example

[Add Damper to Motion Study \(VBA\)](#)  
[Add Damper to Motion Study \(VB.NET\)](#)  
[Add Damper to Motion Study \(C#\)](#)

### Accessors

[IFeature::GetDefinition](#) and [IFeature::IGetDefintion](#)  
[IMotionStudy::CreateFeature](#)

### Access Diagram

[SimulationDamperFeatureData](#)

### See Also

[ISimulationDamperFeatureData Members](#)  
[SolidWorks.Interop.sldworks Namespace](#)  
[ISimulation3DContactFeatureData Interface](#)  
[ISimulationForceFeatureData Interface](#)  
[ISimulationGravityFeatureData Interface](#)  
[ISimulationLinearSpringFeatureData Interface](#)  
[ISimulationMotorFeatureData Interface](#)  
[ISimulationSpringFeatureData Interface](#)

## ISimulationDamperFeatureData Interface Methods

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) : ISimulationDamperFeatureData Interface

For a list of all members of this type, see [ISimulationDamperFeatureData members](#).

### Public Methods

| Name                                                                                                          | Description                                  |
|---------------------------------------------------------------------------------------------------------------|----------------------------------------------|
|  <a href="#">GetEndPoint</a> | Gets the end points for this damper feature. |
|  <a href="#">SetEndPoint</a> | Sets the end points for this damper feature. |

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## See Also

[ISimulationDamperFeatureData Interface](#)  
[SolidWorks.Interop.sldworks Namespace](#)  
[ISimulation3DContactFeatureData Interface](#)  
[ISimulationForceFeatureData Interface](#)  
[ISimulationGravityFeatureData Interface](#)  
[ISimulationLinearSpringFeatureData Interface](#)  
[ISimulationMotorFeatureData Interface](#)  
[ISimulationSpringFeatureData Interface](#)

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## ISimulationDamperFeatureData Interface Members

[See Also](#) [Properties](#) [Methods](#)

[SolidWorks.Interop.sldworks Namespace](#) : ISimulationDamperFeatureData Interface

The following tables list the members exposed by [ISimulationDamperFeatureData](#).

## Public Properties

| Name                                                                                                                              | Description                                                                                            |
|-----------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------|
|  <a href="#">BaseComponent</a>                   | Gets or sets the base component for this damper feature.                                               |
|  <a href="#">DampingConstant</a>                 | Gets or sets the damping constant value for this damper feature.                                       |
|  <a href="#">ExponentOfDamperForceExpression</a> | Gets or sets the exponent for the functional expression for this damper feature.                       |
|  <a href="#">LoadReferences</a>                  | Gets or sets the load references for this damper feature.                                              |
|  <a href="#">Type</a>                           | Gets the type of damper.<br><b>NOTE:</b> This property is a get-only property. Set is not implemented. |

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## Public Methods

| Name                                                                                                            | Description                                  |
|-----------------------------------------------------------------------------------------------------------------|----------------------------------------------|
|  <a href="#">GetEndPoint</a> | Gets the end points for this damper feature. |
|  <a href="#">SetEndPoint</a> | Sets the end points for this damper feature. |

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## See Also

[ISimulationDamperFeatureData Interface](#)  
[SolidWorks.Interop.sldworks Namespace](#)  
[ISimulation3DContactFeatureData Interface](#)  
[ISimulationForceFeatureData Interface](#)  
[ISimulationGravityFeatureData Interface](#)  
[ISimulationLinearSpringFeatureData Interface](#)  
[ISimulationMotorFeatureData Interface](#)  
[ISimulationSpringFeatureData Interface](#)

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## ISimulationDamperFeatureData Interface Properties

[See Also](#)

[SolidWorks.Interop.sldworks Namespace](#) : ISimulationDamperFeatureData Interface

For a list of all members of this type, see [ISimulationDamperFeatureData members](#).

## Public Properties

| Name                                                                                                                | Description                                                                      |
|---------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------|
|  <a href="#">BaseComponent</a>   | Gets or sets the base component for this damper feature.                         |
|  <a href="#">DampingConstant</a> | Gets or sets the damping constant value for this damper feature.                 |
| <a href="#">ExponentOfDamperForceExpression</a>                                                                     | Gets or sets the exponent for the functional expression for this damper feature. |



[LoadReferences](#)

Gets or sets the load references for this damper feature.



[Type](#)

Gets the type of damper.

**NOTE:** This property is a get-only property. Set is not implemented.

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## See Also

[ISimulationDamperFeatureData Interface](#)  
[SolidWorks.Interop.sldworks Namespace](#)  
[ISimulation3DContactFeatureData Interface](#)  
[ISimulationForceFeatureData Interface](#)  
[ISimulationGravityFeatureData Interface](#)  
[ISimulationLinearSpringFeatureData Interface](#)  
[ISimulationMotorFeatureData Interface](#)  
[ISimulationSpringFeatureData Interface](#)

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## BaseComponent Property (ISimulationDamperFeatureData)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [ISimulationDamperFeatureData Interface](#) : BaseComponent Property (ISimulationDamperFeatureData)

Gets or sets the base component for this damper feature.

## .NET Syntax

Visual Basic (Declaration)

Property BaseComponent As Component2

Visual Basic (Usage)



```
Dim instance As ISimulationDamperFeatureData
Dim value As Component2
```

```
instance.BaseComponent = value
```

```
value = instance.BaseComponent
```

C#

```
Component2 BaseComponent {get; set;}
```

C++/CLI

```
property Component2^ BaseComponent {
    Component2^ get();
    void set ( & Component2^ value);
}
```

**NOTE:** See [Differences Between Unmanaged C++ and C++/CLI Code](#).

### Property Value

Base [component](#)

## Visual Basic for Applications (VBA) Syntax

See [SimulationDamperFeatureData::BaseComponent](#).

## See Also

[ISimulationDamperFeatureData Interface](#)  
[ISimulationDamperFeatureData Members](#)

## Availability

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## DampingConstant Property (ISimulationDamperFeatureData)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [ISimulationDamperFeatureData Interface](#) : DampingConstant Property (ISimulationDamperFeatureData)

Gets or sets the damping constant value for this damper feature.

### .NET Syntax

Visual Basic (Declaration)

Property DampingConstant As System.Double

Visual Basic (Usage)



```
Dim instance As ISimulationDamperFeatureData  
Dim value As System.Double
```

```
instance.DampingConstant = value
```

```
value = instance.DampingConstant
```

C#

```
System.double DampingConstant {get; set;}
```

C++/CLI

```
property System.double DampingConstant {  
    System.double get();  
    void set ( & System.double value);  
}
```

**NOTE:** See [Differences Between Unmanaged C++ and C++/CLI Code](#).

#### Property Value

Damping constant value

### Visual Basic for Applications (VBA) Syntax

See [SimulationDamperFeatureData::DampingConstant](#).

### See Also

[ISimulationDamperFeatureData Interface](#)  
[ISimulationDamperFeatureData Members](#)

### Availability

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## ExponentOfDamperForceExpression Property (ISimulationDamperFeatureData)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [ISimulationDamperFeatureData Interface](#) : ExponentOfDamperForceExpression Property (ISimulationDamperFeatureData)

Gets or sets the exponent for the functional expression for this damper feature.

### .NET Syntax

Visual Basic (Declaration)

Property ExponentOfDamperForceExpression As System.Integer

Visual Basic (Usage)



```
Dim instance As ISimulationDamperFeatureData  
Dim value As System.Integer
```

```
instance.ExponentOfDamperForceExpression = value
value = instance.ExponentOfDamperForceExpression
```

C#

```
System.int ExponentOfDamperForceExpression {get; set;}
```

C++/CLI

```
property System.int ExponentOfDamperForceExpression {
    System.int get();
    void set ( & System.int value);
}
```

**NOTE:** See [Differences Between Unmanaged C++ and C++/CLI Code](#).

#### Property Value

Exponent for the functional expression

### Visual Basic for Applications (VBA) Syntax

See [SimulationDamperFeatureData::ExponentOfDamperForceExpression](#).

### See Also

[ISimulationDamperFeatureData Interface](#)  
[ISimulationDamperFeatureData Members](#)

### Availability

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## GetEndPoints Method (ISimulationDamperFeatureData)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [ISimulationDamperFeatureData Interface](#) : GetEndPoints Method (ISimulationDamperFeatureData)

Gets the end points for this damper feature.

### .NET Syntax

Visual Basic (Declaration)

```
Function GetEndPoints( _
    ByRef PDirDisp1 As System.Object, _
    ByRef PDirDisp2 As System.Object, _
    ByRef Type1 As System.Integer, _
    ByRef Type2 As System.Integer _
) As System.Boolean
```

Visual Basic (Usage)



```
Dim instance As ISimulationDamperFeatureData
Dim PDirDisp1 As System.Object
Dim PDirDisp2 As System.Object
Dim Type1 As System.Integer
Dim Type2 As System.Integer
Dim value As System.Boolean
```

```
value = instance.GetEndPoints(PDirDisp1, PDirDisp2, Type1, Type2)
```

C#

```
System.bool GetEndPoints(
    out System.object PDirDisp1,
    out System.object PDirDisp2,
```

```

        out System.int Type1,
        out System.int Type2
    )

```

C++/CLI

```

System.bool GetEndPoints(
& [Out] System.Object^ PDirDisp1,
& [Out] System.Object^ PDirDisp2,
& [Out] System.int Type1,
& [Out] System.int Type2
)

```

**NOTE:** See [Differences Between Unmanaged C++ and C++/CLI Code](#).

#### Parameters

*PDirDisp1*

Damper end point

*PDirDisp2*

Damper end point

*Type1*

Type of end point as defined in [swSelectType\\_e](#)

*Type2*

Type of end point as defined in [swSelectType\\_e](#)

#### Return Value

True if the operation succeeds, false if it fails

### Visual Basic for Applications (VBA) Syntax

See [SimulationDamperFeatureData::GetEndPoints](#).

### See Also

[ISimulationDamperFeatureData Interface](#)

[ISimulationDamperFeatureData Members](#)

[ISimulationDamperFeatureData::SetEndPoints Method](#)

### Availability

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## LoadReferences Property (ISimulationDamperFeatureData)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [ISimulationDamperFeatureData Interface](#) : LoadReferences Property (ISimulationDamperFeatureData)

Gets or sets the load references for this damper feature.

### .NET Syntax

Visual Basic (Declaration)

```
Property LoadReferences As System.Object
```

Visual Basic (Usage)



```
Dim instance As ISimulationDamperFeatureData
```

```
Dim value As System.Object
```

```
instance.LoadReferences = value
```

```
value = instance.LoadReferences
```

C#

```
System.object LoadReferences {get; set;}
```

C++/CLI



```
property System.Object^ LoadReferences {
    System.Object^ get();
    void set ( & System.Object^ value);
}
```

**NOTE:** See [Differences Between Unmanaged C++ and C++/CLI Code](#).

#### Property Value

Load-bearing [faces](#) or [edges](#)

### Visual Basic for Applications (VBA) Syntax

See [SimulationDamperFeatureData::LoadReferences](#).

### See Also

[ISimulationDamperFeatureData Interface](#)  
[ISimulationDamperFeatureData Members](#)

### Availability

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## SetEndPoints Method (ISimulationDamperFeatureData)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [ISimulationDamperFeatureData Interface](#) : SetEndPoints Method (ISimulationDamperFeatureData)

Sets the end points for this damper feature.

### .NET Syntax

Visual Basic (Declaration)

```
Function SetEndPoints( _
    ByVal PDirDisp1 As System.Object, _
    ByVal PDirDisp2 As System.Object _
) As System.Boolean
```

Visual Basic (Usage)



```
Dim instance As ISimulationDamperFeatureData
Dim PDirDisp1 As System.Object
Dim PDirDisp2 As System.Object
Dim value As System.Boolean

value = instance.SetEndPoints(PDirDisp1, PDirDisp2)
```

C#

```
System.bool SetEndPoints(
    System.object PDirDisp1,
    System.object PDirDisp2
)
```

C++/CLI

```
System.bool SetEndPoints(
    & System.Object^ PDirDisp1,
    & System.Object^ PDirDisp2
)
```

**NOTE:** See [Differences Between Unmanaged C++ and C++/CLI Code](#).

#### Parameters

*PDirDisp1*  
 Damper end point  
*PDirDisp2*  
 Damper end point

## Return Value

True if the operation succeeds, false if it fails

## Visual Basic for Applications (VBA) Syntax

See [SimulationDamperFeatureData::SetEndPoints](#).

## Example

See [ISimulationDamperFeatureData](#) examples.

## See Also

[ISimulationDamperFeatureData Interface](#)  
[ISimulationDamperFeatureData Members](#)  
[ISimulationDamperFeatureData::GetEndPoints Method](#)

## Availability

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## Type Property (ISimulationDamperFeatureData)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [ISimulationDamperFeatureData Interface](#) : Type Property (ISimulationDamperFeatureData)

Gets the type of damper.

**NOTE:** This property is a get-only property. Set is not implemented.

## .NET Syntax

Visual Basic (Declaration)

ReadOnly Property Type As System.Integer

Visual Basic (Usage)



```
Dim instance As ISimulationDamperFeatureData  
Dim value As System.Integer  
  
value = instance.Type
```

C#

```
System.int Type {get;}
```

C++/CLI

```
property System.int Type {  
    System.int get();  
}
```

**NOTE:** See [Differences Between Unmanaged C++ and C++/CLI Code](#).

## Property Value

Type of damper as defined in [swSimulationDamperType\\_e](#)

## Visual Basic for Applications (VBA) Syntax

See [SimulationDamperFeatureData::Type](#).

## Example

See [ISimulationDamperFeatureData](#) examples.

## See Also

[ISimulationDamperFeatureData Interface](#)  
[ISimulationDamperFeatureData Members](#)

## Availability

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